

COMMONWEALTH OF AUSTRALIA

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SENG1050

Web Technologies

Week 9 - Website Design and HCI

Lecture Outline

- ❑ Website Structure
- ❑ Web Design Principles
- ❑ Search Engine Optimisation



Recall from last week: JavaScript

- ❑ Last week, we completed the separation of concerns by splitting off the client-side behaviour using JavaScript.
- ❑ We learned to use JavaScript to add dynamic behaviour to our websites, to validate forms, to manipulate page elements in the DOM, to read and write cookies, and to create visual effects.
- ❑ Today, we will address the issue of design – how to make our websites fast, responsive, and attractive to both humans (website design), and computers (particularly search engines, SEO).

Website Structure



What makes a Good Website?

1. A Good Website is Compelling: *It Provides Value to Your Audience that draws them to your site again and again.*
2. A Good Website Meets Clearly Identified Goals.
3. A Good Website is Easy to Navigate.
4. A Good Website features visual attractiveness.
5. A Good website is trustworthy, safe and secure.



Human Computer Interaction

- ❑ Human Computer Interaction (HCI) refers to the study of interfaces between humans and computers. It is a multi-disciplinary field, covering aspects of interface design, graphics, human psychology, and fundamental communication theory, among others.
- ❑ Developers should consider **how users interact** with computer systems, using specific **interfaces**, what the **system requirements** are, what the **user's goals** will be, and how they can **best achieve** them.
- ❑ HCI involves analysing the requirements of a given system, identifying users and their goals; developing models that represent system states, user behaviour and predicted actions; prototyping interfaces, testing, interpreting test results, and iterating on design until the interface requirements have been met.



Planning your Website using HCI principles

1. Identify the goals of your website
2. Identify your audience
3. Create a list of content that both achieves your goals and is of interest to your audience
4. Storyboard your website
 - a. Divide this list into pages
 - b. Draw a chart of links between pages
 - c. Create a navigation scheme that links these pages together in a simple consistent way



Conventional Website Organization

- ❑ Linear
- ❑ Hierarchical
- ❑ Mixed structure

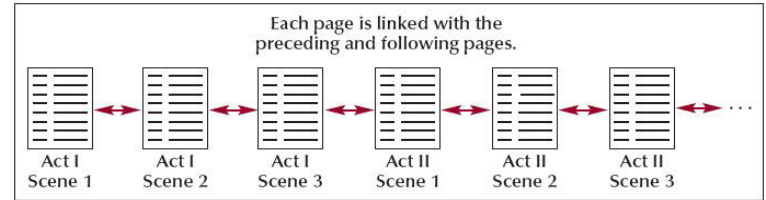


Linear Structures

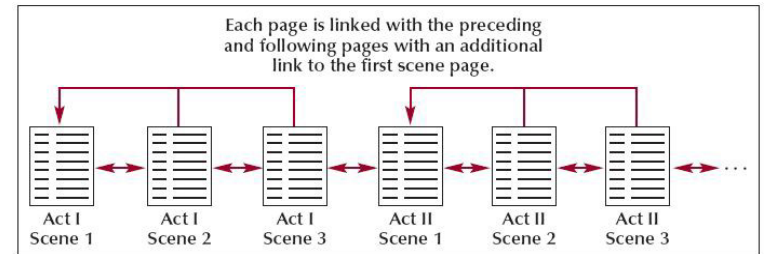
- ❑ In a **linear structure**, each page is linked with the pages that follow and precede it in an ordered chain.
- ❑ **Linear structure** works best for Web pages with a clearly defined order.
- ❑ In an **augmented linear structure**, each page contains an additional link back to an opening page.

Linear Structures

A linear structure



An augmented linear structure



Linear Organisation

- ❑ A series of pages that provide a tutorial, tour, or presentation.
- ❑ Sequential viewing.

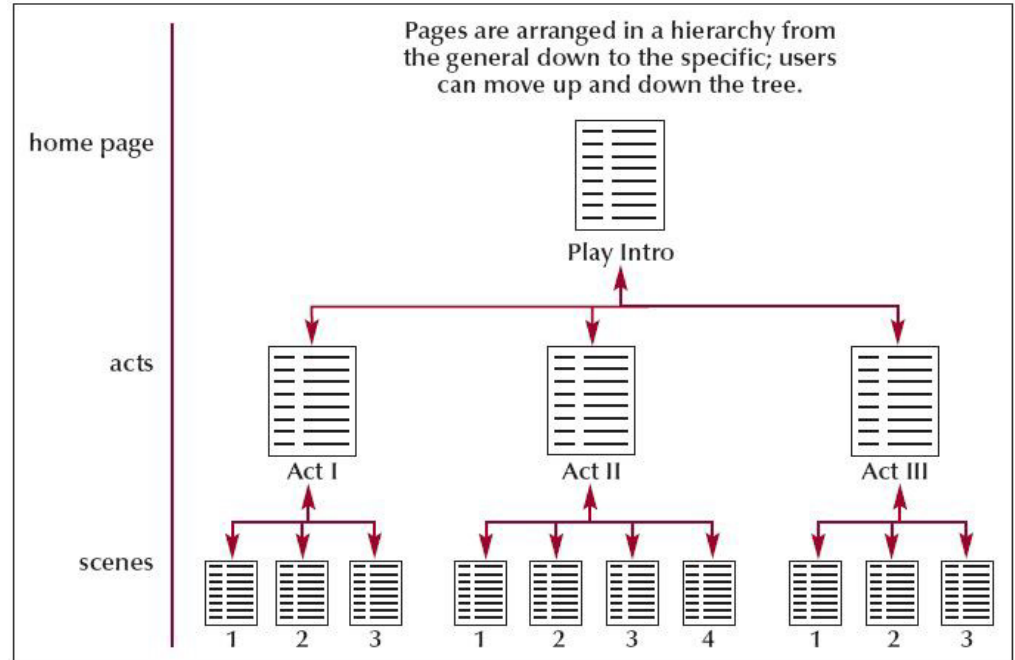




Hierarchical Structures

- ❑ In the **hierarchical structure**, the pages are linked going from the home page down to more specific pages.
- ❑ Users can easily move from general to specific and back again.
- ❑ Within this structure, a user can move quickly to a specific scene within the page, bypassing the need to move through each scene in the play.

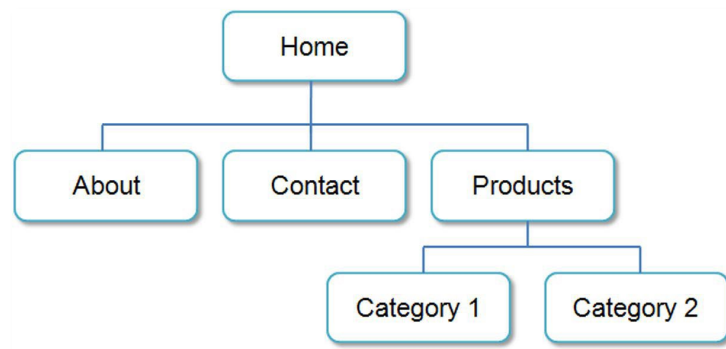
Hierarchical Structures





Hierarchical Organisation

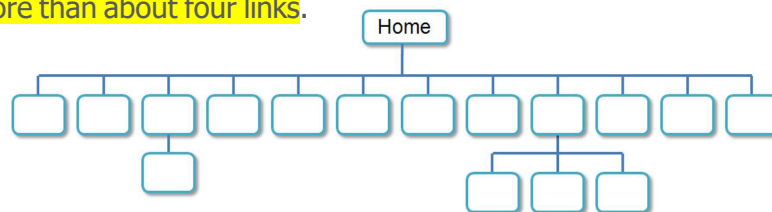
- ❑ A clearly defined home page.
- ❑ Navigation links to major site sections.
- ❑ Often used for commercial and corporate websites.



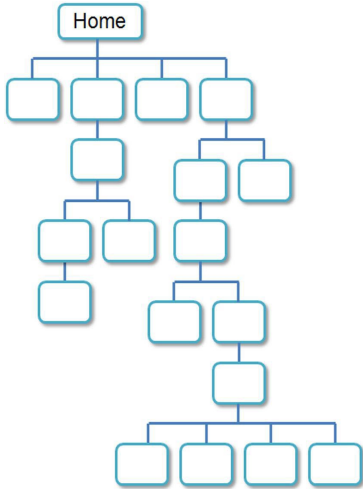


Hierarchical & Shallow

- ❑ Be careful that the organisation is not too shallow.
- ❑ Too many choices → a confusing and less usable web site.
- ❑ Information chunking
 - Research by Nelson Cowan: *adults typically can keep about four items or chunks of items in short-term memory* (<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2864034/>)
- ❑ Be aware of the number of major navigation links.
- ❑ Try group navigation links visually into groups with **no more than about four links.**



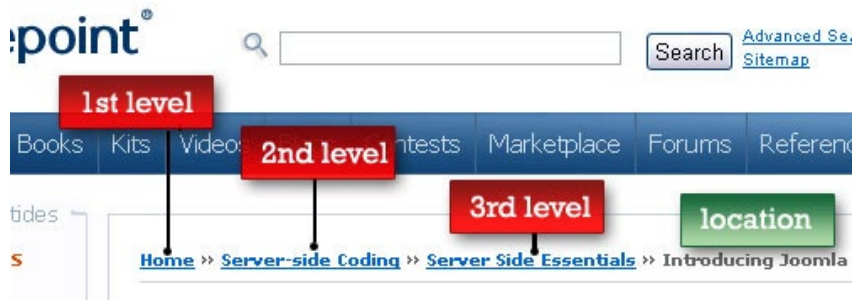
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Breadcrumbs

- ❑ One way to address the issues present in a deep hierarchical organisation is by using 'breadcrumbs'.
- ❑ Breadcrumbs are a navigational aid that shows the website visitor where they are in the website hierarchy relative to the 'root' (the homepage).
- ❑ Each breadcrumb in the chain links to the root page of each level in the hierarchy, giving visitors a quick way to go back to previously visited pages.

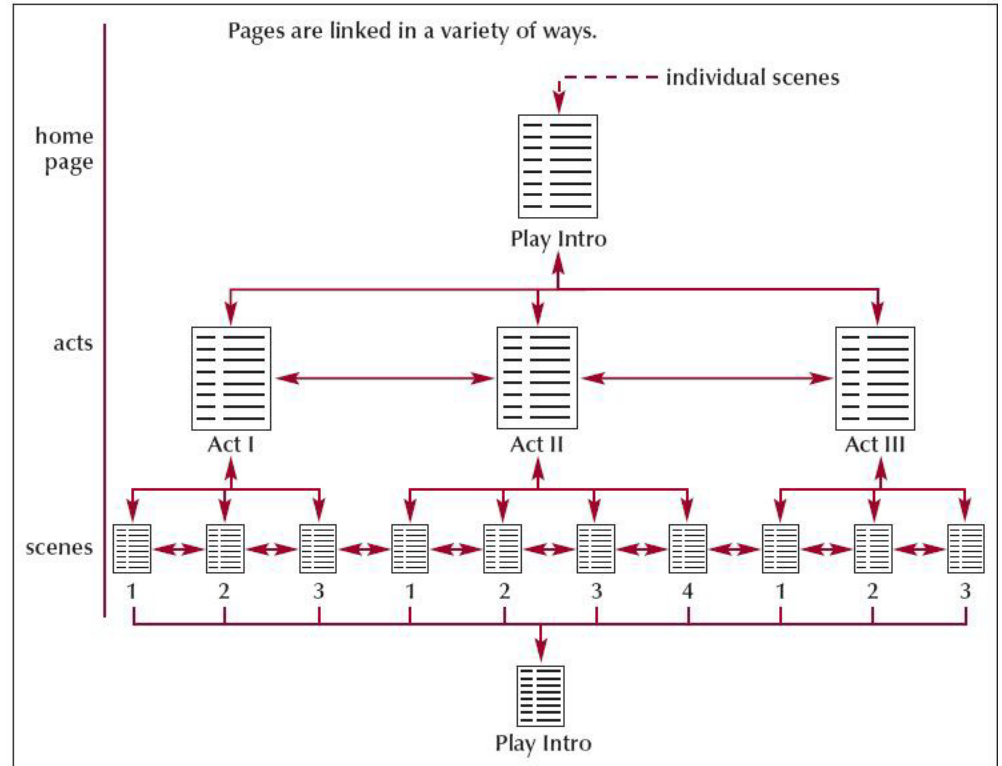




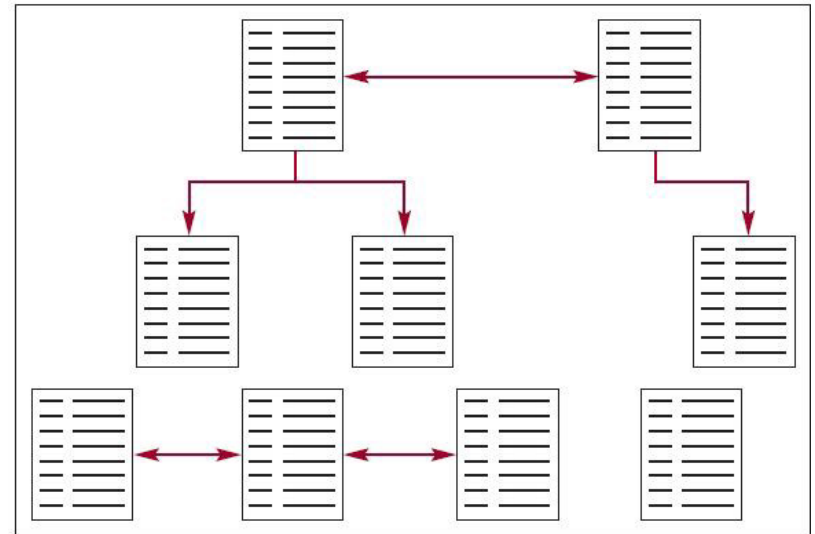
Mixed Structures

- ❑ As Websites become larger and more complex, you often need to use a combination of several different structures.
- ❑ The overall form can be **hierarchical**, allowing the user to move from general to specific; however, the links also allow users to move through the site in a **linear** fashion.

Mixed Structures

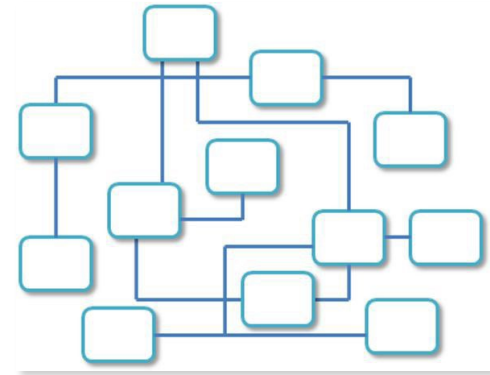


Website with No Coherent Structure



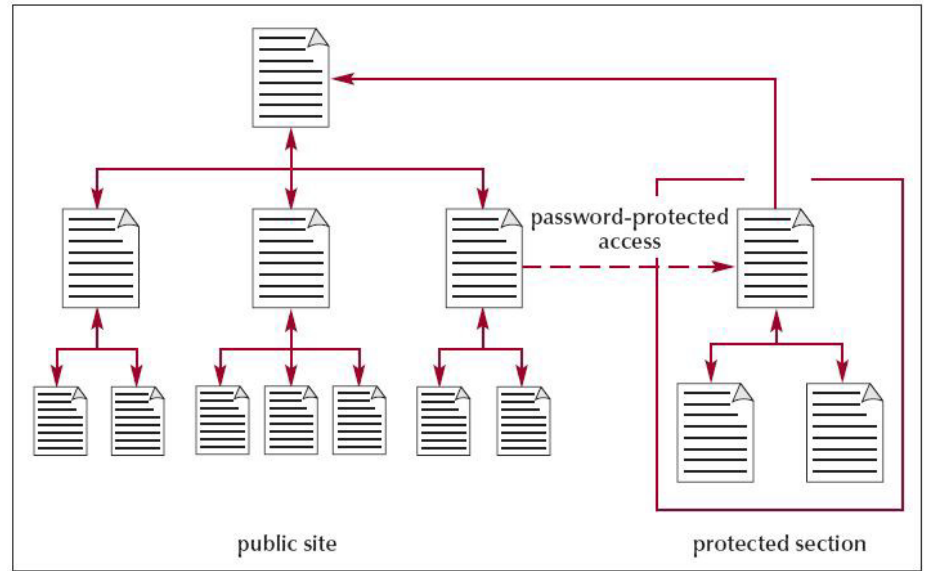
Website with No Coherent Structure

- ❑ Sometimes called “Web” Organisation
- ❑ Usually there is no clear path through the site
- ❑ May be used with artistic or concept sites
- ❑ Not typically used for commercial sites.



Protected Structures

- ❑ Sections of most commercial Web sites are off-limits except to subscribers and registered customers.





Sitemap

Sitemap

A diagram or page containing an outline of the entire website and its contents.

- ❑ A *sitemap* is a file where you provide information about the pages, videos, and other files on your site, and the relationships between them.
- ❑ Search engines read this file to crawl your site more efficiently.

You might need a sitemap if:

- ❑ Your website is very large (more than 500 pages)
- ❑ Your site is new and has few external links to it.

But generally, you don't need a sitemap if your website is small (less than 500 pages) or is comprehensively linked internally.

Which of the following website structures would be most suitable for a website selling products under multiple categories.

Augmented linear

0%

Hierarchical

0%

Linear

0%

Breadcrumbs

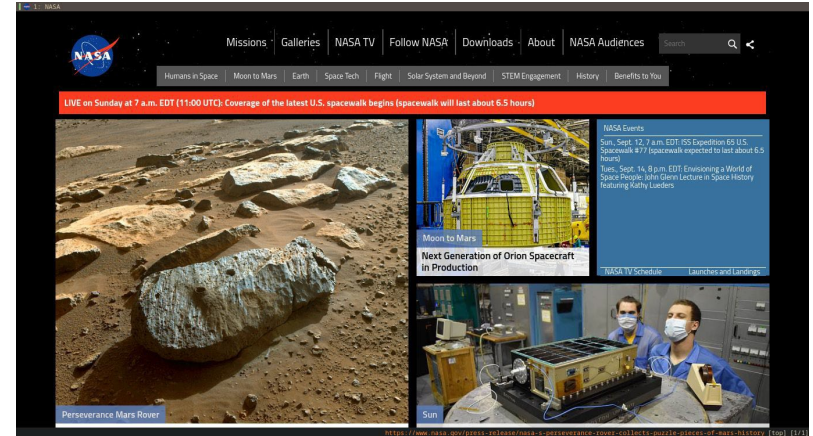
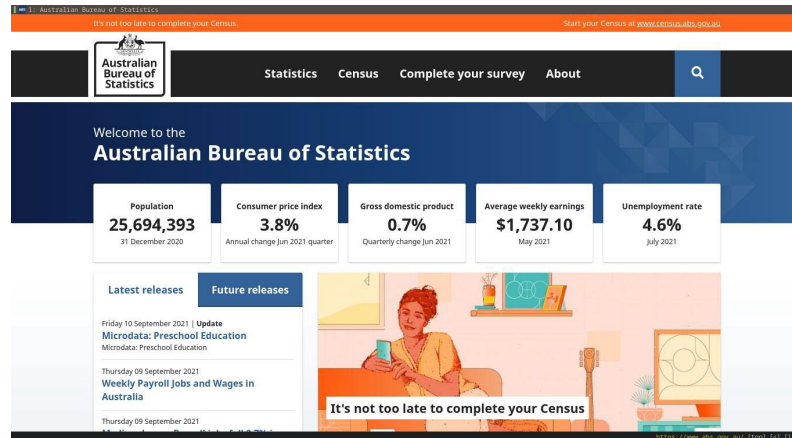
0%



Web Design Principles

Main Principle

- ❑ Overall Design is related to the site's purpose.
- ❑ Consider the target audience of these sites.



<https://www.nasa.gov/>

<https://www.abs.gov.au/>



Web Design Principles

- ❑ Repetition
 - Repeat visual elements throughout design.
- ❑ Contrast
 - Add visual excitement and draw attention.
- ❑ Proximity
 - Group related items.
- ❑ Alignment
 - Align elements to create visual unity.



Design to Provide for Accessibility

“The power of the Web is in its universality. Access by everyone regardless of disability is an essential aspect.”

- *Tim Berners-Lee*

- ❑ Who benefits from increased accessibility?
 - A person with a physical disability
 - A person using a slow Internet connection
 - A person using an old, outdated computer
- ❑ Legal Requirement: Accessible to people with disability:
<https://www.w3.org/WAI/intro/people-use-web/stories>



Design to Provide for Accessibility

- ❑ Content must be **perceivable**.
 - Content is easy to see or hear.
 - Any graphics or multimedia must have a corresponding text content.
- ❑ Interface must be **operable**.
 - Navigation and forms should be easily operable with keyboard/mouse.
 - No flashing content.
- ❑ **Understandable**
 - Content is easy to read and has consistent organization.
- ❑ Robust
 - Support current and future user agents (browsers), including assistive technologies.



Design to Provide for Accessibility

- ❑ Set 'alt' attributes for all images – screen readers will read these even when the image loads.
- ❑ Set the page language in the 'html' element – e.g. `<html lang="en">`
- ❑ Headings should be in descending order without skipping levels (i.e. don't jump from h2 to h5). Screen readers and crawlers can more easily understand the structure of the page.
- ❑ Form inputs should all have labels to assist screen readers.

Writing for the Web

- ❑ Avoid long blocks of text
- ❑ Use bullet points
- ❑ Use headings and subheadings
- ❑ Use short paragraphs

“People don't really read web pages; they scan them”

- *Jakob Nielsen*



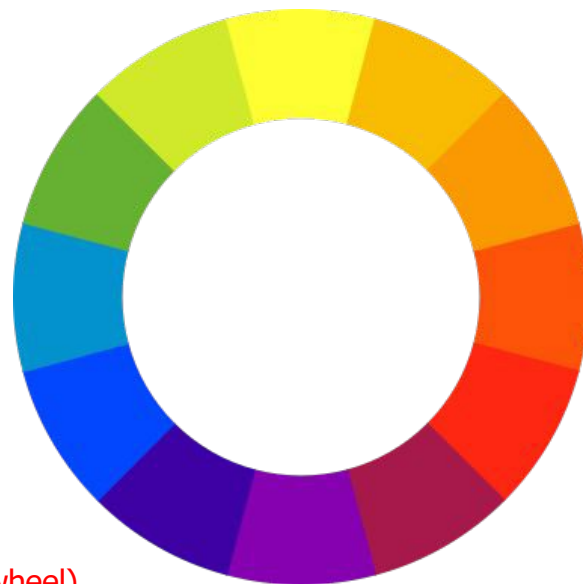
Design “Easy to Read” Text

- ❑ Use common “browser safe” fonts:
 - Arial, Helvetica, Verdana, Times New Roman
- ❑ Use appropriate text size.
- ❑ Use strong contrast between text & background.
- ❑ Use columns instead of wide areas of horizontal text.
- ❑ Carefully choose text in hyperlinks.
 - Avoid “click here”.
 - Hyperlink key words or phrases, not entire sentences.
- ❑ Chek yur spellin (Check your spelling)
- ❑ Match the reading level and knowledge of your audience.



Use Colour Theory

- ❑ Colour Theory:
 - the study of color and its use in design
- ❑ Colour Wheel
 - Primary Colours
 - Secondary Colours
 - Tertiary Colours
- ❑ Use *colour change* to show *status change*
- ❑ Don't use too many colours
 - Use complementary color pairs (opposites on the colour wheel) for best contrast.
- ❑ Use colour coding consistently



https://commons.wikimedia.org/wiki/File:BYR_color_wheel.svg

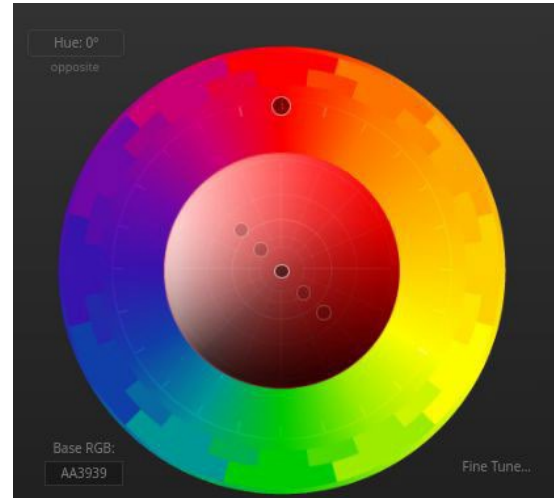
Verify Sufficient Contrast

When you choose colours for text and background, sufficient contrast is needed so that the text is easy to read.

- ❑ Use one of the following online tools to verify contrast:
 - <http://webaim.org/resources/contrastchecker/>
 - http://snook.ca/technical/colour_contrast/colour.html
 - <http://juicystudio.com/services/luminositycontrastratio.php>

Color Scheme Resources

- <https://coolors.co/>
- <https://paletton.com/>
- <https://colorshemedesigner.com/csd-3.5/>
- <https://color.adobe.com/create/color-wheel>
- <http://colormind.io/>
- <https://www.colorsfire.com/>

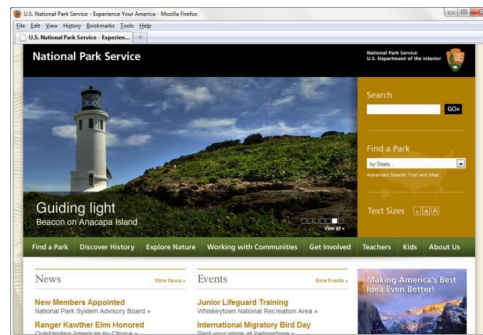


<https://paletton.com/>

Color & Target Audience



Appealing to Kids & Preteens



Appealing to Everyone



Appealing to Young Adults



Appealing to Older Adults



Error Message Guidelines

- ❑ *Speak the user's language*
- ❑ Give *constructive advice* for recovering from the error
- ❑ Indicate negative consequences of the error (e.g., possibly corrupted files)
- ❑ Give an *audible or visual cue*
- ❑ Don't make the user feel guilty!

Email address

Enter your email address

This field has an error

BAD

GOOD

Name

Name is required.

Your full name

Email

Not a valid email address.

A valid email address

Password

Must be 8 characters or more

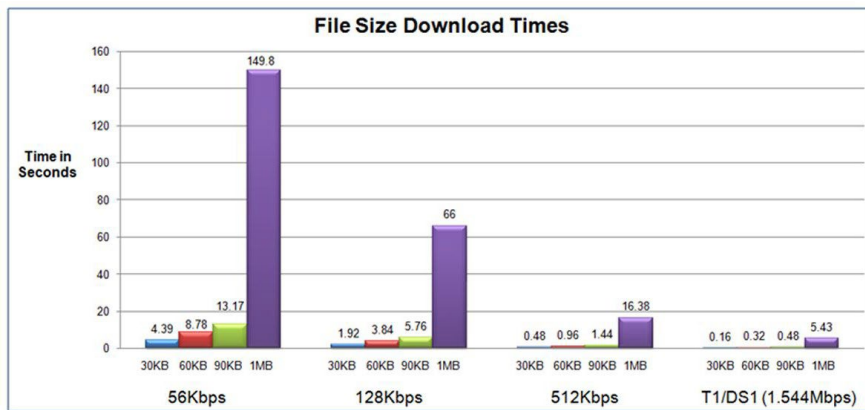
Submit





Web Page Load Time

- Webpages that take longer than about 3 seconds to load lose an average of 50% of traffic.
- Watch the load time of your pages
 - <http://gtmetrix.com/>
- Try to limit web page document and associated media to under 60K on the home page
- Reuse images





Web Page Load Time

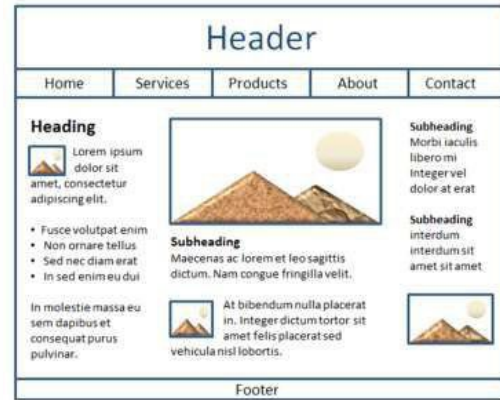
- ❑ Statistically, there are three main causes of slow page load:
 - ❑ Images – Use properly sized images, compress all images (next week), provide multiple versions depending on the screen size available (covered in week 5).
 - ❑ Lazy-load images (``) to only load when scrolled on-screen.
 - ❑ JavaScript – Scripts will block page rendering when they are reached in a file. Split scripts that aren't needed right away into a separate JS file, and use the 'defer' attribute (e.g., `<script defer src="less-important.js"></script>`)
 - ❑ CSS – CSS will also block page rendering until it has been processed. Only above-the-fold CSS should block rendering, split other CSS into a separate file and use 'preload' (e.g., `<link rel="preload" href="style.css" as="style" onload="this.onload=null;this.rel='stylesheet'"><noscript><link rel="stylesheet" href="style.css"></noscript>`)

Browsers & Screen Resolution

- ❑ Test with multiple browsers.
- ❑ Google Chrome, Mozilla Firefox, Apple Safari
- ❑ Test at various screen resolutions.
- ❑ Most widely used: 1920x1080 (23%), 1366x768 (18%), 1536x864 (11%), 1440x900 (6%),
- ❑ Design to look good at various screen resolutions.
 - Centered page content.
 - Set to either a fixed or percentage width.

Wireframe

- ❑ A sketch of blueprint of a web page
- ❑ Shows the structure of the basic page elements, including:
 - ❑ Logo
 - ❑ Navigation
 - ❑ Content
 - ❑ Footer



Web Page Design Page Layout (1)

- ❑ Place the most important information "above the fold" (i.e. content visible immediately on page load)
- ❑ Use adequate "white" or blank space
- ❑ Use an interesting page layout

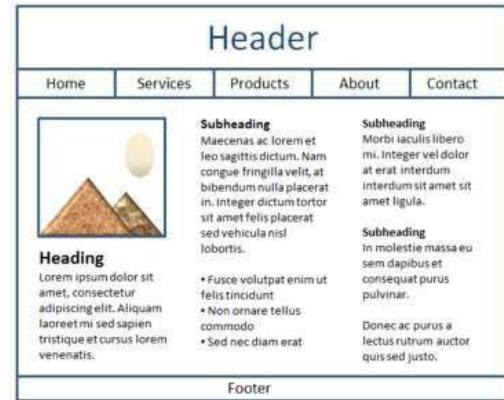


This is usable, but a little boring. See the next slide for improvements in page layout.

Web Page Design Page Layout (2)

Better

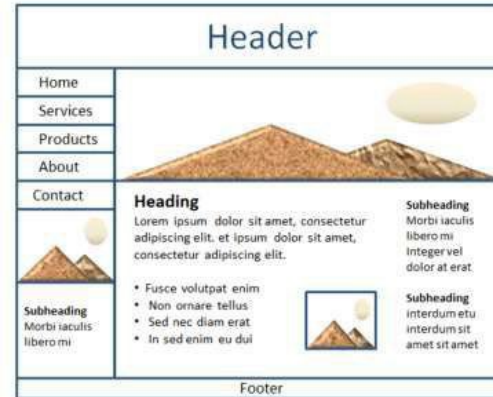
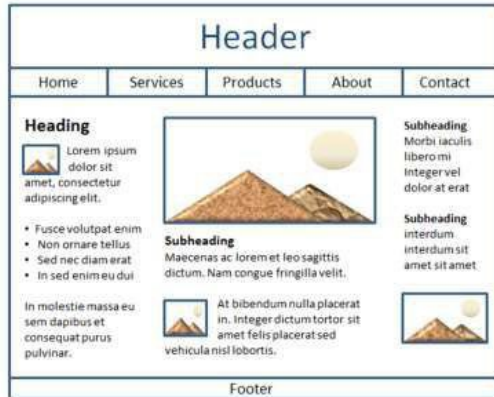
Columns make the page more interesting and it's easier to read this way.



Web Page Design Page Layout (3)

Best So Far

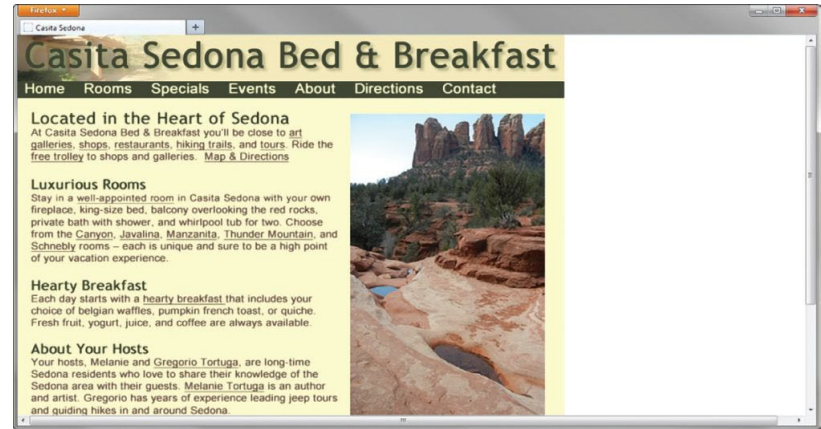
Columns of different widths interspersed with graphics and headings create the most interesting, easy to read page.



Page Layout Design Techniques

Fixed Layout

- ❑ AKA rigid or “ice” design
- ❑ Fixed-width often at left margin
- ❑ Elements do not change when browser is resized



Page Layout Design Techniques

Jello Design

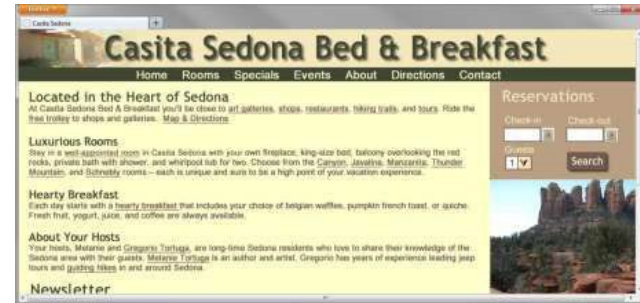
- ❑ Content is centered
 - ❑ Fixed width OR
 - ❑ %width
- ❑ More interesting



Page Layout Design Techniques

Fluid Layout

- ❑ AKA “liquid” design
- ❑ Expands to fill the browser at all resolutions.
- ❑ The most common layout these days





Media queries

- ❑ The same layout rarely works on all screen sizes. Smaller screens in particular usually require a different layout to be used.
- ❑ CSS rules can be optionally applied based on conditions set in **media queries**.
- ❑ The syntax is: `@media <conditions> { selector {property: value; ...} }`
- ❑ Examples:
 - `@media (min-width: 600px) {body {margin: 0px}}`
 - `@media (max-height: 400px) {main {background-image: url("background-mobile.png")}}`
 - `@media (orientation: portrait) {header {padding: 0px}}`
 - `@media (hover: none) {div.link {font-size: 1.2em}}` // **Good for detecting mobile devices and tablets vs desktops; (hover: hover) generally detects desktops.**



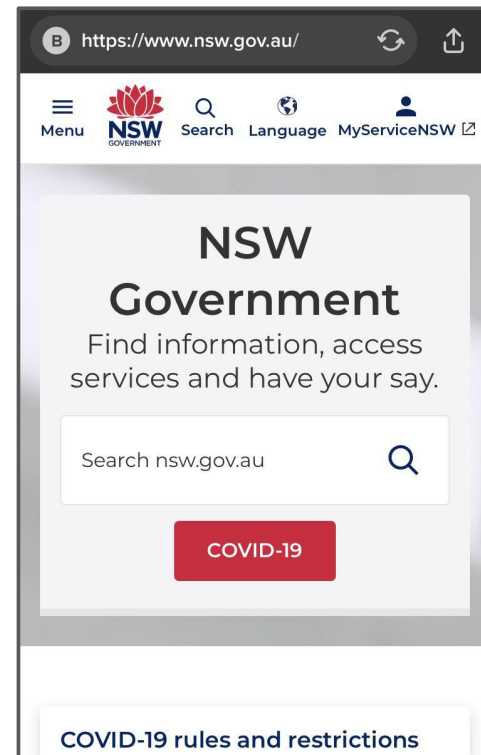
Design for the Mobile Web

- ❑ More people access websites using mobile devices than with desktop computers
- ❑ >68% of all website visits came from mobile devices in 2020 (Proficient Inc.)
- ❑ Three Approaches:
 - ❑ Separate .mobi mobile site
 - ❑ Host the mobile site within your current domain
 - ❑ Configure your current website for mobile display using **responsive web** design techniques
- ❑ Add the meta element: `<meta name="viewport" content="width=device-width, initial-scale=1.0">`
This indicates to search engines that your site is mobile-friendly.



Mobile Design Quick Checklist

- ☐ Small(er) screen size
- ☐ Bandwidth issues??
- ☐ Single-column layout
- ☐ Maximize contrast
- ☐ Optimize images for mobile display
- ☐ Descriptive alternate text for images
- ☐ Avoid display of non-essential content
 - ☐ Provide easy access to your website's features
 - ☐ Prominent hyperlinks or search button



Responsive Web Design

- ❑ Term coined by Ethan Marcotte, noted web developer:
<http://www.alistapart.com/articles/responsive-web-design>
- ❑ Progressively enhancing a web page for different viewing contexts (such as smartphones and tablets) through the use of coding techniques, including flexible layouts and media queries.
- ❑ Examples: <http://www.mediaqueri.es>
- ❑ Chromium-based browsers such as Google Chrome and Microsoft Edge provide a tool in the Inspector called **Lighthouse**, which checks many of these elements for you – required for Assessment 3 (Web-Based Assignment 2).

What Not to Do

- ❏ <http://www.webpagesthatsuck.com/>
 - Some brilliant examples of rubbish
- ❏ <http://www.ratz.com/featuresbad.html>
 - A “Thou shalt not” list
- ❏ <http://www.angelfire.com/super/badwebs/>
 - World’s Worst Website
- ❏ <https://arngren.net/>

Which of the following is NOT a principle for writing text on the web?

Text should be structured using headings and subheadings.

0%

Using wide areas of text going from left and right (according to natural reading order) rather than colum...

0%

Related items should be grouped using visually related bullet points.

0%

Short, concise paragraphs are preferred to long blocks of text.

0%

It is preferred to use fonts are common and available to most clients.

0%



Which of the following methods CANNOT be used to decrease the load time of images on a webpage?

Implementing lazy-loading for images that are below-the-fold.

0%

Compressing images using a modern image format such as webp.

0%

Using the srcset and sizes attributes on an img element to provide the image at different resolutions.

0%

Preloading images using the 'preload' attribute of a picture element.

0%



Which of the following media queries with a CSS rule would apply a larger font-size to smaller devices (such as mobile phones)?

@media (min-width: 200px) {{body: font-size: 120%;}}

0%

@media (max-width: 900px) {{body: font-size: 110%;}}

0%

@media (max-width: 2000px) {{body: font-size: 20%;}}

0%

@media (min-width: 600px) {{body: font-size: 80%;}}

0%



Search Engine Optimisation



Why is Internet Marketing important?

It gives companies the ability to:

- ❑ Drive high quality customers to their website
- ❑ Increase sales leads from customers looking for their products and services
- ❑ Build their brand online by communicating marketing messages to their target audience
- ❑ Establish two-way communication with the audience
- ❑ Target a global audience via international search engines
- ❑ Provide personalised experience to their customers

<https://www.linkedin.com/pulse/7-reasons-behind-importance-internet-marketing-blog-media-group-s>



Internet Marketing

Key Internet marketing strategies

- ❑ Search engine optimisation (SEO)
 - no cost per click
- ❑ Search engine marketing (SEM)
 - cost per click

Search Phrase

Google

best bikes for kids



All Shopping Images News Videos More

Tools

Number of search results

About 355,000,000 results (0.68 seconds)

SEO vs SEA

Paid search results

Organic search results

Ad - <https://www.vulyplay.com/kids/bikes> - 1800 676 345
Tough Kids Bikes - The New Vuly Bikes
Durable Innovative **Kids Bikes** by Vuly®. Buy online and save. Free Delivery Australia Wide. Vuly® **kids bikes**. Buy online and save. Sale ending soon! Free Delivery Australia Wide. Innovative Technology. Sale Ends Soon! We Deliver Nationwide. Huge Discounts!
★★★★★ Rating for vulyplay.com: 4.6 - 986 reviews
Huge Sale! Prices Slashes - Contact Us - We Delivery Nationwide

Ad - <https://www.reviewrubble.com/kids-best-kids-bike> -
Top 5 Kids Bikes - September 2021 - reviewrubble.com
Sept 2021 Review: Highest Quality, Most Durable **Kids Bikes**. From \$149. Balance, Training & Mountain **Bikes**. Don't Overpay. Trusted Trampoline. **Best For Kids**. Safest Trampoline. Trampolines on Sale. Brands: Vuly, Vuly Play, Flare.
Bikes For 7-12 Year Olds - Bikes for 2 - 6 Year Olds - Safest Kids Bikes

Ad - <https://www.99bikes.com.au/> -
Bikes For Children - For Ages 2 - 12 Years Old
99 **Bikes** stocks **children's bikes** in popular sizes like 12 Inch, 16 Inch, 20 Inch & 24 Inch

Ad - <https://www.bicyclensw.org.au/> - (02) 9704 0800
Buying a Bike for Children - Guide For Buying A Kids Bike
Congratulations on deciding to get the **child** in your life riding. Be sure to buy the right size...

<https://www.bicycling.com/bikes-gear/best-kids-bikes> -
The 10 Best Kids Bikes You Can Buy Right Now - Bicycling
28 Dec 2020 — Like Strider, Woom specializes in **kids' bikes**. With 12-inch wheels and a saddle that can be set low to the ground, the Woom 1 balance **bike** can ...

<https://www.twowheelingtots.com/best-kids-bikes-the-...> -
10 Best Kids Bikes: Our Favorite Brands and Where to Buy Them
23 Apr 2021 — The woom 2 (14") and woom 3 (16") **bikes** are the perfect first pedal **bikes**. They are the **LIGHTEST kids bikes** on the U.S. market, which has a HUGE ...

Ads - Shop best bikes for kids



Vuly Kids
Bike (20"...)
\$450.00
Vuly
★★★★★ (20)



Vuly Kids
Bike (20"...)
\$450.00
Vuly
★★★★★ (20)



Vuly Kids
Bike (20"...)
\$450.00
Vuly
★★★★★ (8)



Vuly Kids
Bike (20"...)
\$450.00
Vuly
★★★★★ (4)



Goldcross
Kids...
rebel
\$179.99
★★★★★ (40)



KTM
FACTORY...
Procycles
\$1,430.00
★★★★★ (40)

Search Engine Optimisation

What is SEO?

SEO is the act of modifying a website to increase its ranking in **organic (vs paid)**, crawler-based listings of search engines

"Spiders read only text, nothing else"

How do organic search listings work?

- ❑ A **spider** or **crawler** is a component of a Search Engine (SE) that gathers listings by automatically "crawling" the web
- ❑ The spider follows links to web pages, makes copies of the pages and stores them in the SE's index
- ❑ Based on this data, the SE then **indexes** the pages and **ranks** the websites
- ❑ Major SEs that index pages using spiders: Google, Bing, DuckDuckGo



Search Engine Optimisation

- ❑ Domain Name
- ❑ TITLE tags
- ❑ Meta Description
- ❑ Keywords used in texts
 - H1, H2, H3...
- ❑ Alt Tags
- ❑ Links
 - Anchor text in links
 - Internal and external links
- ❑ SEs

Search Engine Optimisation - Domain Name

Domain name strategies

- ❑ domain names are traffic magnets
 - All the authority your brand has built will be tied to the domain name.
 - But more than 523 million domain names have been registered.
- ❑ choose a domain name that will increase your search engine ranking. How?
 - simple, short, no hyphens, no numbers
- ❑ use keywords, common words, advertising terms, product names
- ❑ choose a keyword that is important for your business

Search Engine Optimisation - Title Tags

Important in optimising rankings

- ❑ The first thing that a search engine displays on a search return
- ❑ Every single web page must have its own title tag
- ❑ Create accurate, unique page titles
- ❑ Use keywords in your titles (see <https://www.ebay.com.au/>)
- ❑ Don't choose a title that has no relation to the content of the page
- ❑ Don't stuff unneeded keywords in your title tags



Search Engine Optimisation - Title Tags

Your site's Title MUST grab attention, create curiosity, evoke emotion to get the click. Don't just use your company's or website's name.

Spend the same amount of time when writing your Title as you would if you were writing a Pay Per Click (PPC) ad. Because:
No clicks = No conversions

Ad · <https://www.bunnings.com.au/> ▼

Bunnings Warehouse - Shop Our Wide Range Online

View Our Range Of Trusted Brands At Warehouse Prices. Save When You D.I.Y. At **Bunnings**. Lowest Prices Has Always Been A Policy At **Bunnings**...It Is Behind Everything We Do. D.I.Y. advice. Over 320 stores. Great service. Expert advice.

📍 Penrith · 8 locations nearby

Bunnings Marketplace

Greater Choice In One Place.
Delivering You Even More Value.

Bunnings D.I.Y. Advice

View A Wide Range of D.I.Y. Videos & Articles From Bunnings.

Gift Cards

Buy A Gift Card Online.
Do More With Warehouse Prices.

Make It Yours Series

1 House. 13 Style Creators. 1 Epic Makeover. Watch Online Now.

<https://www.bunnings.com.au/> ▼

Australia DIY, Garden & Hardware Store - Bunnings Australia

Find store information, opening times, services, tools and more. Come visit us today.

Results from bunnings.com.au





Search Engine Optimisation - Meta Description

```
<meta name="description"
content="Brandon's Baseball Cards provides a large selection of
vintage and modern baseball cards for sale. We also offer daily
baseball news and events.">
```

- ❑ `<meta>` description gives search engines a summary of what the page is about.
- ❑ A page's description `<meta>` tag might be a sentence or two or even a short paragraph.
- ❑ But the `<meta>` description might not always be used.
- ❑ SEs may choose to use a relevant section of your page's visible text if it does a good job of matching up with a user's query.



Search Engine Optimisation - Keywords

❑ important in optimising rankings

- ❑ Keywords are words that appear the most in a page
- ❑ The spider chooses the appropriate keywords for each page, then sends them back to its SE
- ❑ Your web site will then be indexed based on your keywords
- ❑ Don't repeat keywords excessively, or use irrelevant keywords – search engines can detect this, and will penalise your website as a result.

Keywords

- ❑ Can be key phrases or a single keyword
 - ❑ Do not use common words e.g., 'the' 'and' 'of'
 - spiders ignore them
 - ❑ Write keyword-rich text and focus on the quality of the content
- ❑ Balance keyword-rich and readability
- ❑ Always have text in your page
 - at least 100 words

Search Engine Optimisation - Links

Linking strategies

- ❑ The more inbound links the higher the SE ranking.
- ❑ If the site linking to you is already indexed, spiders will also receive your site.
- ❑ Quality of inbound links is critical, esp from authority websites.
- ❑ How to increase links:
 - Good content
 - Good outbound links
 - Target a list of sites from which you can request inbound links

“Link relevancy is critical in getting your site indexed by search engines”

“A small number of inbound links from high-quality, relevant sites is more valuable than many links from low-traffic, irrelevant sites.”

Authority website: A website that is trusted by its users, the industry it operates in, other websites and search engines.

Search Engine Optimisation - SEs

- ❑ In most cases, you do not need to submit your website for indexing
 - The search engines will automatically index your website in due time.
- ❑ But sometimes, your website might not be indexed
 - One possible reason is that there are no external links to your website
 - Or the search engine hasn't crawled your new website yet.
- ❑ In this case, you can submit your url for indexing

- ❑ Google
<https://www.google.com/webmasters/tools/submit-url>
- ❑ Bing
<http://www.bing.com/toolbox/webmaster>
- ❑ Yandex
<https://passport.yandex.com>
- ❑ Entireweb
http://www.entireweb.com/free_submission/

Search Engine Optimisation

PDF files

- ❑ Search engines can read and index PDF files.
- ❑ But you might need to optimize PDF files for SEO
 - Reduce size
 - Keyword-rich file name
 - Text-based pdf not image based
 - Meta description and good content structure
 - Make PDF mobile friendly

Dynamic pages

- ❑ Search engines will index content on pages using ASP, CGI or other dynamic languages.
- ❑ To ensure all the dynamic content is crawled:
 - Avoid using a high number of parameters
 - Use sitemaps to help search engines find your pages.

Search Engine Optimisation

SEO - what is NOT recommended

❑ *Image only sites*

- Spiders do not pick up images

❑ *Image maps*

- Spiders cannot read image maps.
- Do not use them on your home page or critical pages.

❑ *Frames*

- Only one page can be titled (titling is critical in search rankings)
- If the spider cannot read the complete page (because of the frames), it will not be indexed properly.
- Some spiders may not even read a frames website.

❑ *Password protected pages*

- spiders cannot enter password protected pages



Search Engine Optimisation

To summarise

- ❑ Critical elements
 - domain name, links, keywords, title, meta description, alt tags, submitting your website to SEs
- ❑ No-nos
 - image-only sites, image maps, frames, password protected pages

Monitoring Search Listings

- ❑ Once your website is launched, some time may be required before your site appears in search listings.
- ❑ It is important to keep track of which optimisation efforts are working (esp keywords)
 - Then fine tune your optimisation and keywords over time.
- ❑ Web analytics software/services can automatically analyse the traffic to your website and provide helpful insights.
- ❑ Most web host providers allow free access to traffic logs.
- ❑ Other services: Google Analytics, Alexa, SEMRush



Google Analytics

Google Analytics is a powerful and free web analytics tool that provides insights such as:

- ❑ Audience
 - Geography, browser, etc
- ❑ Traffic Sources
 - Referring sites, keywords, and AdWords
- ❑ Content
 - Landing pages, paths through the site, exit pages
- ❑ Conversions
 - Track business objectives



Search engine marketing

What is search engine marketing (SEM)?

SEM is the act of marketing a website via search engines by purchasing paid listings

What are paid listings?

These are listings that search engines sell to advertisers, usually through **paid placement** or **paid inclusion** programs. In contrast, organic listings are not sold.



Search engine marketing

There are two major types of paid listings: paid inclusions and paid placements

Paid inclusions

- ❑ Advertising programs where pages are guaranteed to be included in a search engine's index in exchange for payment
- ❑ no guaranteed ranking
- ❑ payment made on a *Pay Per Click (PPC)* basis
- ❑ eg Looksmart: *a directory that lists pages and sites, not based on position but based on relevance*
- ❑ Advertisers pay to be included in the directory on a PPC basis or per-url fee basis with no guarantee of specific placement



Search engine marketing

Paid placements

- ❑ Advertising programs where listings are guaranteed to appear in organic listings
 - the higher the fee, the higher the ranking
- ❑ e.g., sponsored links and Google's Adwords
- ❑ can be purchased from a portal or a search network
- ❑ search networks are often set up in an auction environment where keywords and phrases are associated with a pay-per-click (PPC) fee.
- ❑ Google Ads, Amazon Advertising and Microsoft Advertising (formerly Bing) are the largest paid placement search networks

Which of the following is NOT a good way of optimising a webpage for search engines?

Using a short, descriptive domain name with a specific keyword that represents the purpose of the site.

0%

Including a meta description for each webpage that describes the purpose and content of the webpage.

0%

Adding high-quality inbound links to your website by including a modified version with extra links in embedded iframes.

0%

Making a mobile-friendly layout for your website to use when accessed by mobile devices.

0%





Summary

- ❑ We have looked at how to design a website that is fast, responsive, and appealing to both human and computer audiences using HCI and SEO principles.
- ❑ We have learned different ways to organise our website to help users easily navigate and find the content they are looking for.
- ❑ Unfortunately, larger websites still have problems; even after all of the optimisations we have discussed, they can be very slow to load.
- ❑ We need to look at compressing our website, to deliver web content in the smallest file size possible to speed up page load and increase traffic to our website – next week.

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